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RAJARATA UNIVERSITY OF SRI LANKA
FACULTY OF APPLIED SCIENCES

Bachelor of Science in Applied Sciences

Second year Semester I Examination – August / September 2024

CHE 2205 – INORGANIC CHEMISTRY

Answer all questions. **Time: Two (02) hours.**

Mass of electron	m_e	=	9.11×10^{-31} kg
Mass of proton	m_p	=	1.672×10^{-27} kg
Mass of neutron	m_n	=	1.675×10^{-27} kg
Avogadro number	N_A	=	6.022×10^{23} mol^{-1}
Universal gas constant	R	=	8.314 J mol^{-1} K^{-1}
Planck constant	h	=	6.63×10^{-34} J s
Speed of light	c	=	3.00×10^{8} m s^{-1}
1 atomic mass unit (amu)	1 amu	=	1.66×10^{-27} kg
	1 eV	=	1.602×10^{-19} J

The use of a non-programmable calculator is permitted.

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1. a) Name the following complexes:

   (i) $[Pt(en)_2Cl_2]Cl_2$

   (ii) $[Cr(NH_3)_3(H_2O)_3]Cl_3$

   (10 marks)

   b) Write the formulae for each of the following compound.

   (i) tetrahydroxidozincate(II) ion

(ii) potassium diaaminetetrachloridochromate(III)

(10 marks)

c) Indicate the coordination number and the oxidation number for the central metal atom in each of the following coordination compounds:

(i) $[Pt(H_2O)_2Br_2]$

(ii) $[Pt(NH_3)(py)(Cl)(Br)]$; (py = pyridine, C_5H_5N)

(iii) $[Ni(H_2O)_4Cl_2]$

(iv) $[Fe(en)_2(CN)_2]^+$

(16 marks)

d) Draw the crystal field diagrams for $[Fe(NO_2)_6]^{4-}$ and $[FeF_6]^{3-}$. State whether each complex is high spin or low spin, paramagnetic or diamagnetic. Calculate the spin only magnetic momentum for each compound and compare Δ_{oct} to P (pairing energy) for each complex.

(20 marks)

e) Predict giving reasons the effect of Jahn-Teller distortion of the following complexes; CuF_2 , $CuBr_2$, $Ni(NH_3)_6]^{2+}$ and $[Ni(H_2O)_6]^{2+}$.

(20 marks)

f) Give five differences between crystalline solid and amorphous solid. Name two substances in each type of solid.

(18 marks)

g) Tungsten crystallizes in the body-centered cubic unit cell. If the edge of the unit cell is 316.5 pm, what is the radius of the tungsten atom?

(16 marks)

h) The density of copper metal is 8.95 g cm^{-3}. If the radius of copper atom is 127.8 pm, is the copper unit cell simple cubic, body centered cubic or face-centered cubic? (Atomic mass of Cu = 63.54 g mol^{-1} and NA = 6.02×10^{23} mol^{-1}).

(18 marks)

(I) Find the wavelength of X-rays in nm for NaCl crystals if the distance between layers (d) = 282 pm, angle of incidence (θ) = 23°, and the order of reflection (n) = 1.

(16 marks)

(j) (i) Define the term isotopes and isobars. Write down the balanced nuclear reaction when a radioactive nucleus ($^{238}_{92}U$) decays one alpha particle and two beta particles.

(ii) The radioactive element $_{88}Ra$ emits three alpha particles in succession. Deduce the resulting element.

(16 marks)

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2. a) i. If Δ_0 is the octahedral crystal field splitting energy. Deduce the CFSE for $Fe(CN)_6]^{4-}$?

ii. Explain the violet colour of $[Ti(H_2O)_6]^{3+}$ complex on the basis of the crystal field theory?

(16 marks)

b) i. $[Fe(CN)_6]^{3-}$ is a common complex of iron. Give the oxidation state and electron configuration for iron in the complex.

ii. Draw a crystal field diagram showing the orbital energies and ground state electron configuration in $Fe(CN)_6]^{3-}$. Deduce the magnetic properties of the complex.

iii. The complex $[Fe(NH_3)_6]^{3+}$ is observed to be much more strongly affected by a magnetic field than $Fe(CN)_6]^{3-}$. Draw crystal field diagram showing the ground state electron configuration in $[Fe(NH_3)_6]^{3+}$.

(34 marks)

c) i. The wavelength of light observed by $[Fe(NH_3)_6]^{3+}$ to be shorter or longer than that observed by $[Fe(CN)_6]^{3-}$. Explain.

ii. Predict the type of ligand, number of unpaired electrons, the magnetic moment of 25 °C for each of the following: $[Ru(NH)6]^{3+}$, $[Cr(NH_3)_6]^{4-}$ and $[EuCl_6]^{4}$.

(30 marks)

3. a) i. Find the number of atoms in bcc and fcc unit cell arrangements.

ii. Calculate the packing efficiency for bcc and fcc lattice structures.

(24 marks)

b) i. Define the following terms in relation to crystalline solids : Unit cell and coordination number. Give one example in each case.

ii. The unit cell of an element of atomic mass 108 amu and density 10.5 g cm^{-3} is a cube with edge length, 409 pm. Find the type of unit cell of the crystal.

(16 marks)

c) Consider a metal with an FCC structure and an atomic weight of 92.9. When monochromatic X-radiation having a wavelength of 0.1028 nm is focused on the crystal, the angle of diffraction (2θ) for the (311) set of planes in this metal occurs at 71.2 degrees (for the first order reflection n=1). Calculate the;

i. Interplanar spacing for this set of planes.

ii. Lattice parameter for this metal.

iii. Density of the metal (units of g/cm^3).

(40 marks)

4. a) Complete the following nuclear reactions.

i. ${}^{7}_{3}Li + {}^{1}_{1}H \rightarrow$ ______

ii. ${}^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{137}_{52}A + {}^{97}_{40}B +$ ______ and identify A nd B

iii. ${}^{82}_{34}Se \rightarrow 2\,{}^{0}_{-1}e +$ ______

iv. ${}^{27}_{13}Al +$ ______ $\rightarrow {}^{28}_{13}Al + \gamma$

(20 marks)

b) i. Calculate the binding energy and binding energy per nucleon (in MeV) of a nitrogen nucleus. (atomic mass of nitrogen nucleus – 14.00307 amu)

ii. Deduce the half-life of a radioactive isotope if a 500 g sample decays to 62.5 g in 24.3 hours.

(35 marks)

c) i. Calculate the time taken for a 40 g sample of ${}^{131}I$, decay to $\frac{1}{100}$ of its original mass. (half-life = 8.040 days).

ii. Give three uses of radioactivity.

(25 marks)

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